1.

a) (2 pts.) The vector wave field $\mathbf{E}(z,t)$ is given by

$$\mathbf{E}(z,t) = 4\Delta(\frac{t-z/c}{\tau})\,\hat{y}\,\frac{\mathrm{V}}{\mathrm{m}}.$$

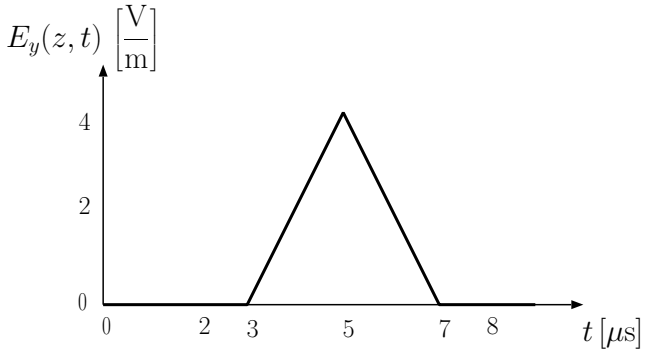
b) (2 pts.) The associated wave field $\mathbf{H}(z,t)$ is

$$\mathbf{H}(z,t) = -\frac{4}{\eta_o}\Delta(\frac{t-z/c}{\tau})\,\hat{x}\,\frac{\mathrm{A}}{\mathrm{m}}.$$

c) (2 pts.) The field E_y given by

$$E_y(z,t)|_{z=1500} = 4\triangle\left(\frac{t-\frac{15\times10^2}{3\times10^8}}{4\times10^{-6}}\right) = 4\triangle\left(\frac{t-5\times10^{-6}}{4\times10^{-6}}\right)\,\frac{\mathrm{V}}{\mathrm{m}},$$

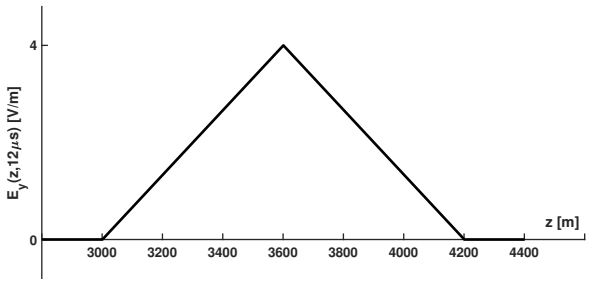
vs t at $z = +1500\,\mathrm{m}$ is plotted in the following figure:



d) (2 pts.)The field E_y given by

$$E_y(z,t)|_{t=12\mu s} = 4\triangle\left(\frac{12\times10^{-6}-\frac{z}{3\times10^8}}{4\times10^{-6}}\right) = 4\triangle\left(\frac{3600-z}{1200}\right)\,\frac{\mathrm{V}}{\mathrm{m}},$$

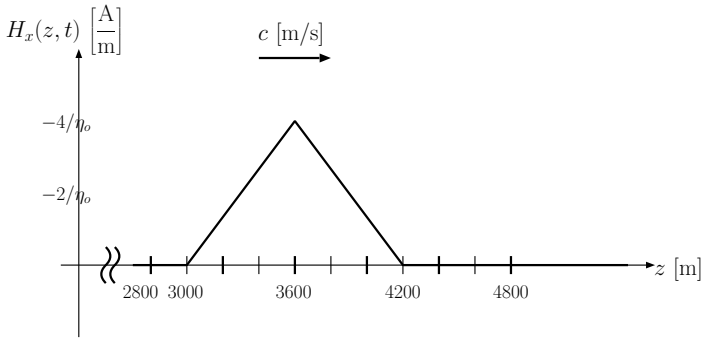
vs z at $t = 12\,\mu s$ is plotted in the following figure:



e) (2 pts.) The field H_x given by

$$H_x(z,t)|_{t=12\,\mu\text{s}} = -\frac{4}{\eta_o}\Delta\left(\frac{12\times10^{-6}-\frac{z}{3\times10^{-8}}}{4\times10^{-6}}\right) = -\frac{4}{\eta_o}\Delta(\frac{3600-z}{1200})\,\frac{\text{A}}{\text{m}},$$

vs x at $t = 12\,\mu$s is plotted in the following figure:



2.

a) (2.5 pts.) The intrinsic impedance of the medium is

$$\eta = \left|\frac{E_x}{H_y}\right| = 120\pi\,\Omega.$$

We have used the orthogonal pair E_x and H_y. The same relation should be valid for the orthogonal pair E_y and H_x.

b) (2.5 pts.) The propagation velocity is given by

$$v = \frac{c}{2} = 1.5\times10^8\,\frac{\text{m}}{\text{s}}.$$

c) (2.5 pts.) If $\epsilon = \epsilon_r\epsilon_o$ and $\mu = \mu_r\mu_o$, then the intrinsic impedance η can take the following form

$$\eta = \sqrt{\frac{\mu}{\epsilon}} = \sqrt{\frac{\mu_r}{\epsilon_r}}\eta_o$$

while the wave propagation velocity v can be written as

$$v = \frac{1}{\sqrt{\epsilon\mu}} = \frac{c}{\sqrt{\epsilon_r\mu_r}}.$$

Using the results of parts (a) and (b) we find that

$$\epsilon_r = \mu_r = 2.$$

d) (2.5 pts.) Finally, since $E_y = -\eta H_x$ (recall that the propagation direction is $\hat{z} = -\hat{y}\times\hat{x}$, hence the minus sign), we have that

$$g(t-\frac{z}{c/2}) = -120\pi\times(\frac{10z}{c}-5t) = 600\pi\times\left(t-\frac{z}{c/2}\right),$$

then

$$g(t) = 600\pi t.$$

3.

a) (2.5 pts.) For the plane wave described by $\mathbf{E}_1 = 4\sin(\omega t - \beta z)\hat{x}\,\frac{\mathrm{V}}{\mathrm{m}}$. The magnetic field $\mathbf{H}$ should satisfy $\mathbf{H} = -\frac{\mathbf{E}\times\hat{\beta}}{\eta}$, where $\hat{\beta}$ is the unit vector parallel to the propagation direction. Then, we can find the expressions for $\mathbf{H}$ field of the given plane wave as

$$\hat{\beta}_1 = \hat{z} \rightarrow \mathbf{H}_1 = \hat{y}\frac{4}{\eta_o}\sin(\omega t - \beta z)\,\frac{\mathrm{A}}{\mathrm{m}}.$$

b) (2.5 pts.) For the plane wave described by $\mathbf{E}_2 = 5\cos(\omega t - \beta z)(\hat{x} - \hat{y})\,\frac{\mathrm{V}}{\mathrm{m}}$. The propagation direction is $\hat{\beta}_2 = \hat{z}$. Thus, the magnetic field $\mathbf{H}_2$ is given by

$$\mathbf{H}_2 = \frac{5}{\eta_o}\cos(\omega t - \beta z)(\hat{y} + \hat{x})\,\frac{\mathrm{A}}{\mathrm{m}}.$$

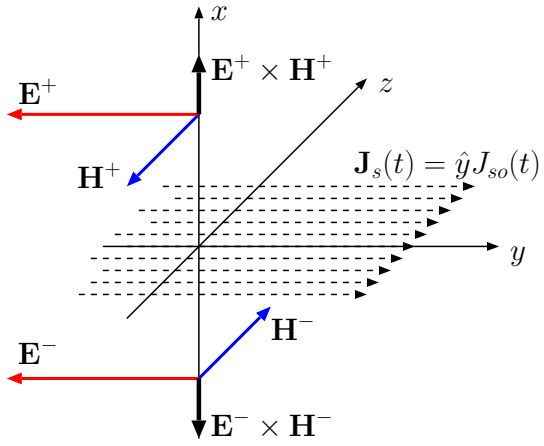
c) (2.5 pts.) For the plane wave described by $\mathbf{H}_3 = 2\cos(\omega t + \beta z + \frac{\pi}{3})\hat{x} - 2\cos(\omega t + \beta z - \frac{\pi}{6})\hat{y}\,\frac{\mathrm{A}}{\mathrm{m}}$. The electric field $\mathbf{E}$ should satisfy $\mathbf{E} = \eta(\mathbf{H}\times\hat{\beta})$ where $\hat{\beta} = -\hat{z}$ in this case. Then, we can find the expressions for the $\mathbf{E}$ field of the given plane wave as

$$\mathbf{E}_3 = \eta_o\left(2\cos(\omega t + \beta z + \frac{\pi}{3})\hat{y} + 2\cos(\omega t + \beta z - \frac{\pi}{6})\hat{x}\right)\,\frac{\mathrm{V}}{\mathrm{m}}.$$

d) (2.5 pts.) For the plane wave described by $\mathbf{H}_4 = 4\cos(\omega t + \beta x)\hat{z} + 4\sin(\omega t + \beta x)\hat{y}\,\frac{\mathrm{A}}{\mathrm{m}}$. The propagation direction is $\hat{\beta}_4 = -\hat{x}$. Thus, the electric field $\mathbf{E}_4$ is given by

$$\mathbf{E}_4 = -4\eta_o\left(\cos(\omega t + \beta x)\hat{y} - \sin(\omega t + \beta x)\hat{z}\right)\,\frac{\mathrm{V}}{\mathrm{m}}.$$

4. The pulse of sheet current $\mathbf{J}_s(t) = \hat{y}4t\mathrm{rect}(\frac{t}{\tau})\,\frac{\mathrm{A}}{\mathrm{m}}$ will produce magnetic and electric fields as shown in the following figure.



The magnetic field is: (it's direction can be verify using the right-hand-rule for Ampere's law $\oint_C \mathbf{H}\cdot d\mathbf{l} = I_C$)

$$\begin{aligned}\mathbf{H}^{\pm}(x,t) &= \mp\frac{1}{2}J_{so}\left(t \mp \frac{x}{c}\right)\hat{z}\,\frac{\mathrm{A}}{\mathrm{m}} \\ &= \mp 2\left(t \mp \frac{x}{c}\right)\mathrm{rect}\left(\frac{t \mp \frac{x}{c}}{\tau}\right)\hat{z}\,\frac{\mathrm{A}}{\mathrm{m}} \quad \text{for } x \gtrless 0,\end{aligned}$$

whereas the electric field is given by

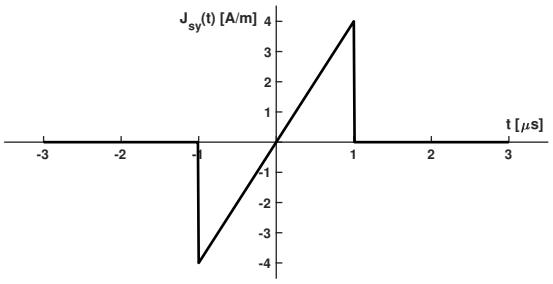
$$\mathbf{E}^{\pm}(x,t) = -\frac{\eta_o}{2} J_{so}\left(t \mp \frac{x}{c}\right)\,\hat{y} = -2\eta_o\left(t \mp \frac{x}{c}\right)\,\text{rect}\left(\frac{t \mp \frac{x}{c}}{\tau}\right)\,\hat{y}\,\frac{\text{V}}{\text{m}} \quad \text{for } x \gtrless 0,$$

where $\eta_o = 120\pi\,\Omega$ is the intrinsic impedance of free-space.

a) (2 pts.) For

$$J_{sy}(t) = 4t\text{rect}(\frac{t}{\tau})\,\text{A/m},$$

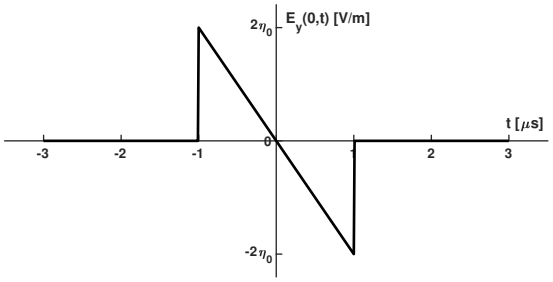
we have the following plot for $-3 < t < 3\,\mu$s :

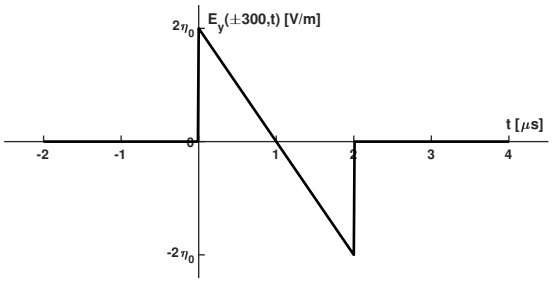


b) (4 pts.) The electric field generated by the current sheet is

$$\mathbf{E}(x,t) = -2\eta_o\left(t \mp \frac{x}{c}\right)\,\text{rect}\left(\frac{t \mp \frac{x}{c}}{\tau}\right)\,\hat{y}\,\frac{\text{V}}{\text{m}},$$

and we can plot it vs $t$at $x = 0$ and $x = \pm 300$ m as:

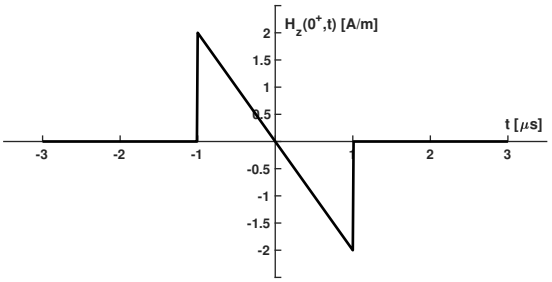


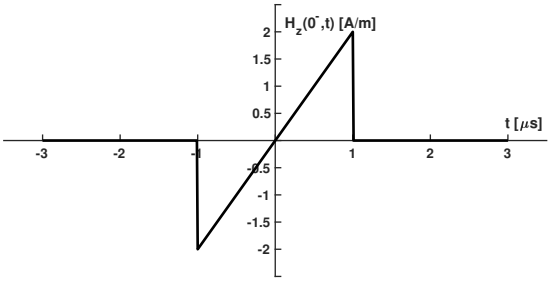


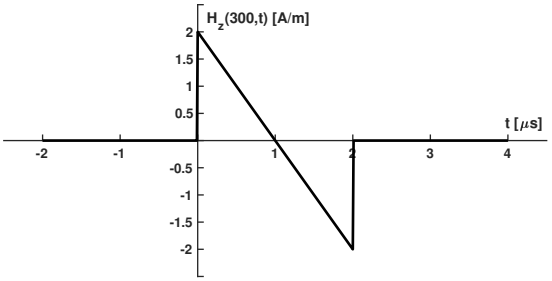
c) (4 pts.) The magnetic field generated by the current sheet is

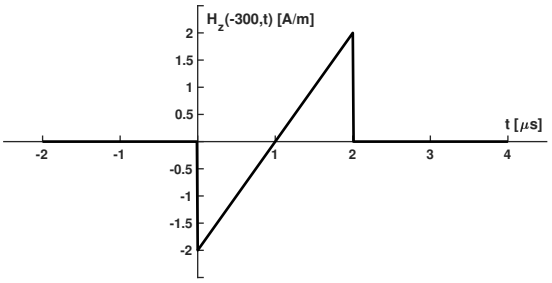
$$\mathbf{H}(x,t) = \mp 2 \left(t \mp \frac{x}{c} \right) \text{rect} \left(\frac{t \mp \frac{x}{c}}{\tau} \right) \hat{z} \, \frac{\text{A}}{\text{m}},$$

and we can plot it vs t at $x = 0^{\mp}$ and $x = \pm 300$ m as:









d) (4 pts) Following the hint given in the problem, we can write

$$\begin{aligned}
-\mathbf{J}_s \cdot \mathbf{E} &= -\left(\hat{y}4t \text{ rect}\left(\frac{t}{\tau}\right)\right) \cdot \left(-2\eta_0 t \text{ rect}\left(\frac{t}{\tau}\right)\hat{y}\right) \\
&= 8\eta_0 t^2 \text{rect}^2\left(\frac{t}{\tau}\right) \frac{\text{W}}{\text{m}^2} \\
&= 8\eta_0 t^2 \text{rect}^2\left(\frac{t}{\tau}\right) \frac{\text{J}}{\text{s}\cdot\text{m}^2}. \\
&= 8\eta_0 t^2 \text{rect}^2\left(\frac{t}{\tau}\right) \frac{\mu\text{J}}{\mu\text{s}\cdot\text{m}^2}
\end{aligned}$$

Then, the TEM wave density energy is

$$\begin{aligned}
\int_{-\infty}^{\infty} -\mathbf{J}_s \cdot \mathbf{E}\, dt &= \int_{-\infty}^{\infty} 8\eta_0 t^2 \text{rect}^2\left(\frac{t}{\tau}\right)\, dt \\
&= \int_{-\tau/2}^{\tau/2} 8\eta_0 t^2\, dt = 8\eta_0 \left.\frac{t^3}{3}\right|_{-\tau/2}^{\tau/2} \\
&= 640\pi \frac{\mu\text{J}}{\text{m}^2} \\
&= 640\pi \times 10^{-6} \frac{\text{J}}{\text{m}^2}.
\end{aligned}$$

e) (4 pts) We need to first calculate the Poynting vector at x = 300 m. This can be done using the results from part (b) and (c) as

$$\begin{aligned}
\mathbf{S} = \mathbf{E} \times \mathbf{H} &= \hat{y} E_y\left(x = 300 \text{ m}, t\right) \times \hat{z} H_z\left(x = 300 \text{ m}, t\right) \\
&= \hat{x} 4\eta_0 \left(t-1\right)^2 \text{rect}^2\left(\frac{t-1}{2}\right) \frac{\mu\text{J}}{\mu\text{s}\cdot\text{m}^2}.
\end{aligned}$$

Then the wave energy radiated will be

$$\begin{aligned}
\int_{-\infty}^{\infty} \mathbf{S}\, dt &= \int_{-\infty}^{\infty} 4\eta_0 \left(t-1\right)^2 \text{rect}^2\left(\frac{t-1}{2}\right)\, dt \\
&= \int_{0}^{2} 4\eta_0 \left(t-1\right)^2\, dt \\
&= 320\pi \times 10^{-6} \frac{\text{J}}{\text{m}^2}.
\end{aligned}$$

for the unit area on the x = 300 m plane.

f) (2 pts) As you may notice, the result in part (e) is exactly half of that in part (d). This is in compliance with the principle of energy conservation since the wave from the current sheet is radiated into positive and negative x directions both with equal magnitudes.